

Tropical Meteorology



Chapter 1: Introduction



Objective

- *To Define Tropical Meteorology*
- *To Know The Climatic boundaries of the tropics*
- *To See Main Characteristics Of The Tropical Weather And Climate*

Tropical Meteorology ?

Meaning and Scope of Tropical Meteorology

- ❑ Tropical meteorology is the study of atmospheric processes occurring in the tropics, the region of the Earth surrounding the equator.
- ❑ Tropical meteorology deals about tropical atmosphere including
 - Thunderstorms and lightning
 - Tropical cyclones (hurricanes, typhoons)
 - Monsoon
 - Madden-Julian Oscillation
 - Easterly jets etc...

Tropical Meteorology?

- ✦ Tropical meteorology also encompasses dealing with important energy sources and sinks like surplus radiation, latent heat, sensible heat, evapotranspiration, and ocean heat storage that drive the abovementioned events.
- ✦ A study of tropical meteorology must also consider tropical atmosphere-ocean coupling.

Three Main Ways To Define The Tropics

- Geographical
- Astronomical
- Dynamical

Geographical Definition of the Tropics

- The geographical definition of the tropics refers to the region of the Earth located between 23.5° North and 23.5° South latitude.
- It is bounded by:
 - Tropic of Cancer (23.5°N)
 - Tropic of Capricorn (23.5°S)
- This definition is based on *fixed latitude lines shown on maps and globes.*
- It defines the tropical zone as a permanent belt around the equator.

Astronomical Definition of the Tropics

- The astronomical definition of the tropics is *based on the apparent movement of the Sun during the year.*
- The tropics are the regions where the Sun can be directly overhead (at the zenith) at least once a year.
- The northernmost and southernmost points where this happens are:
 - Tropic of Cancer
 - Tropic of Capricorn
- These boundaries are determined by the 23.5° tilt of the Earth's axis, which controls the maximum solar declination during the solstices.

Solar Declination and Apparent Movement

Solar declination is the latitude at which the Sun is directly overhead at noon.

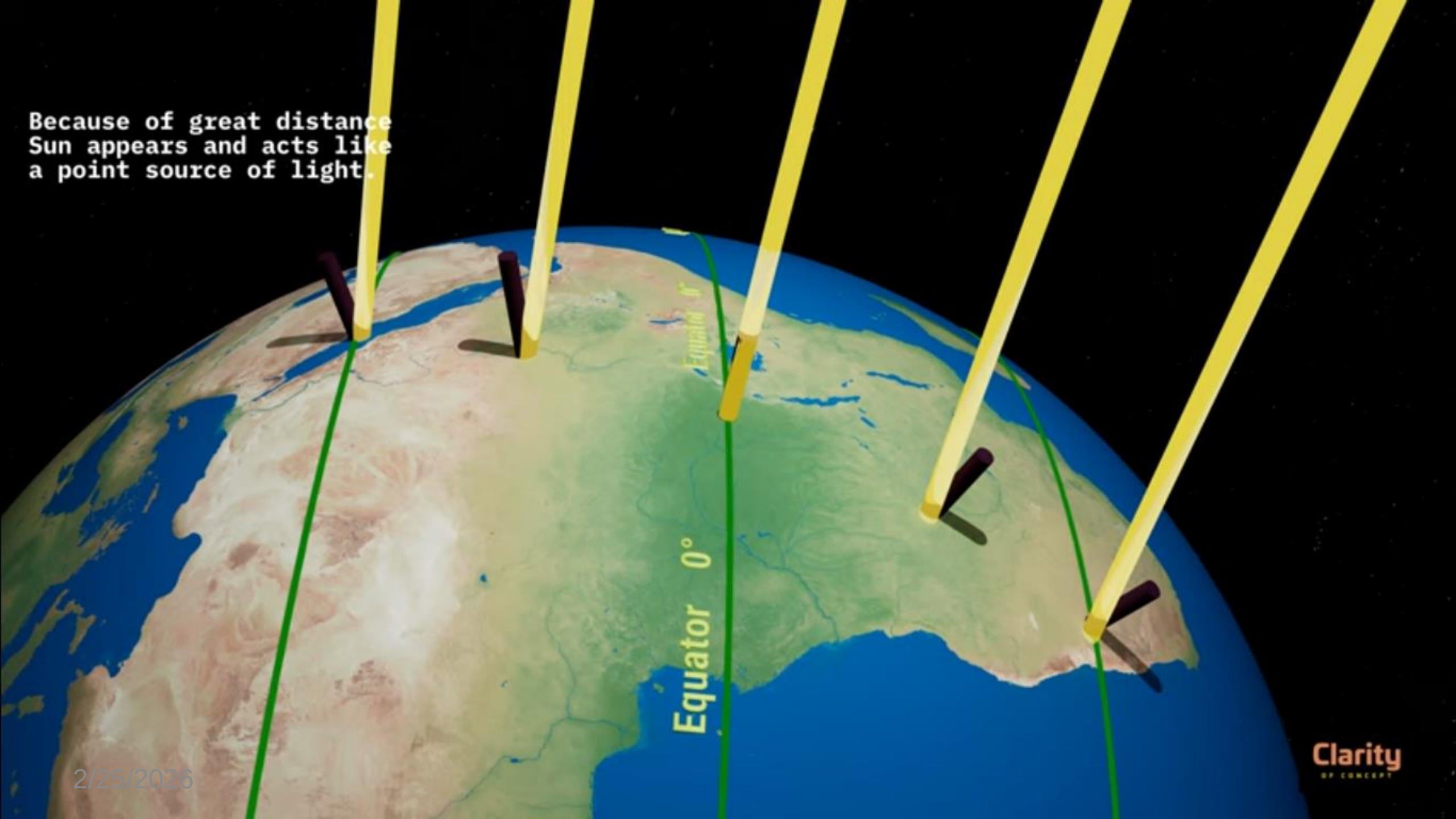
During the year:

- Around June 21 (Northern Hemisphere summer solstice), the Sun is directly overhead at 23.5°N .
- Around December 21 (Southern Hemisphere summer solstice), the Sun is directly overhead at 23.5°S .
- During the equinoxes (March and September), the Sun is directly overhead at the Equator (0°).

Thus, the Sun's vertical rays oscillate between 23.5°N and 23.5°S every year.

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This north-south shift is called the annual apparent movement of the Sun.

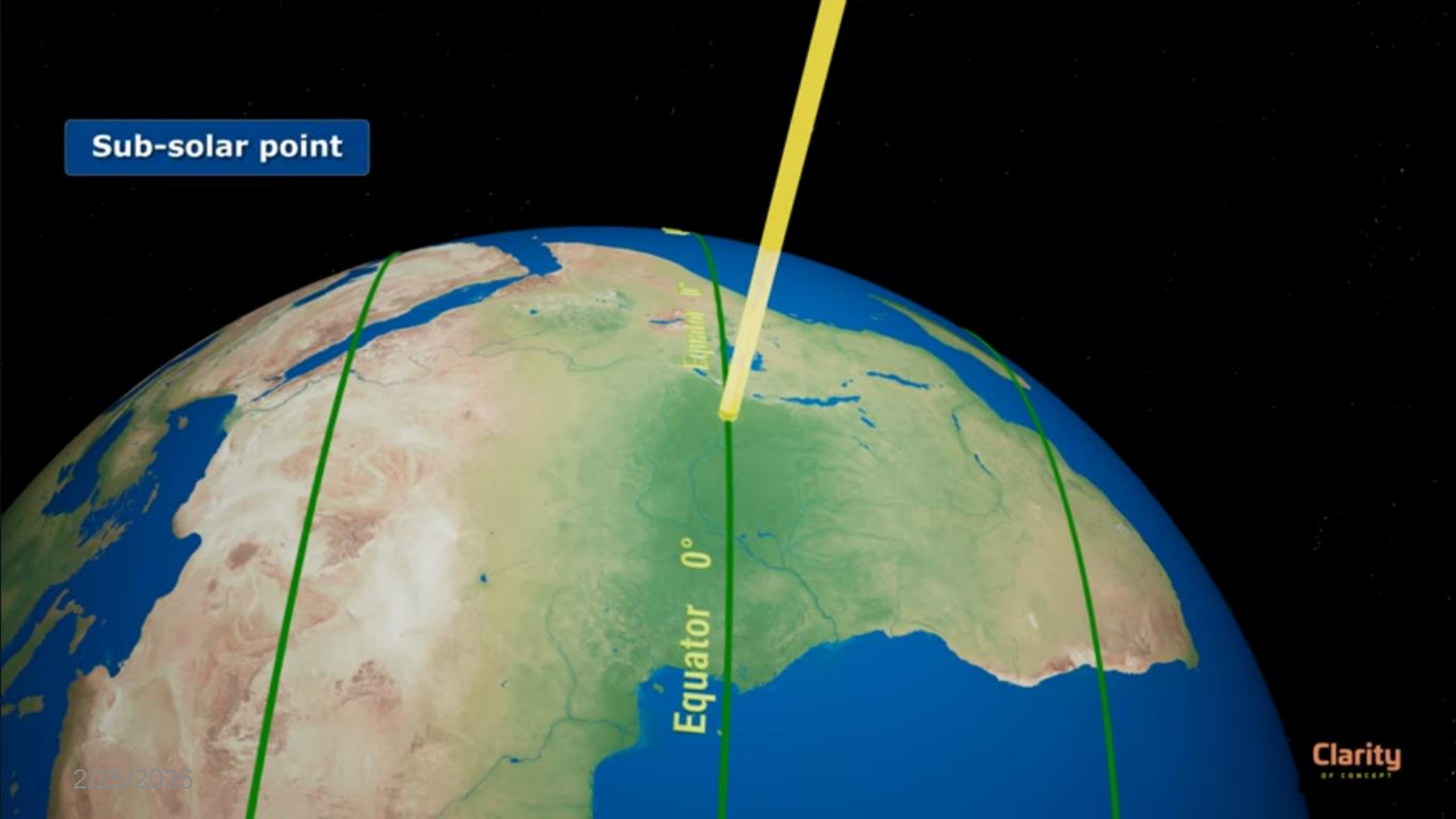
Because of great distance
Sun appears and acts like
a point source of light.



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Clarity
OF CONCEPT

Sub-solar point



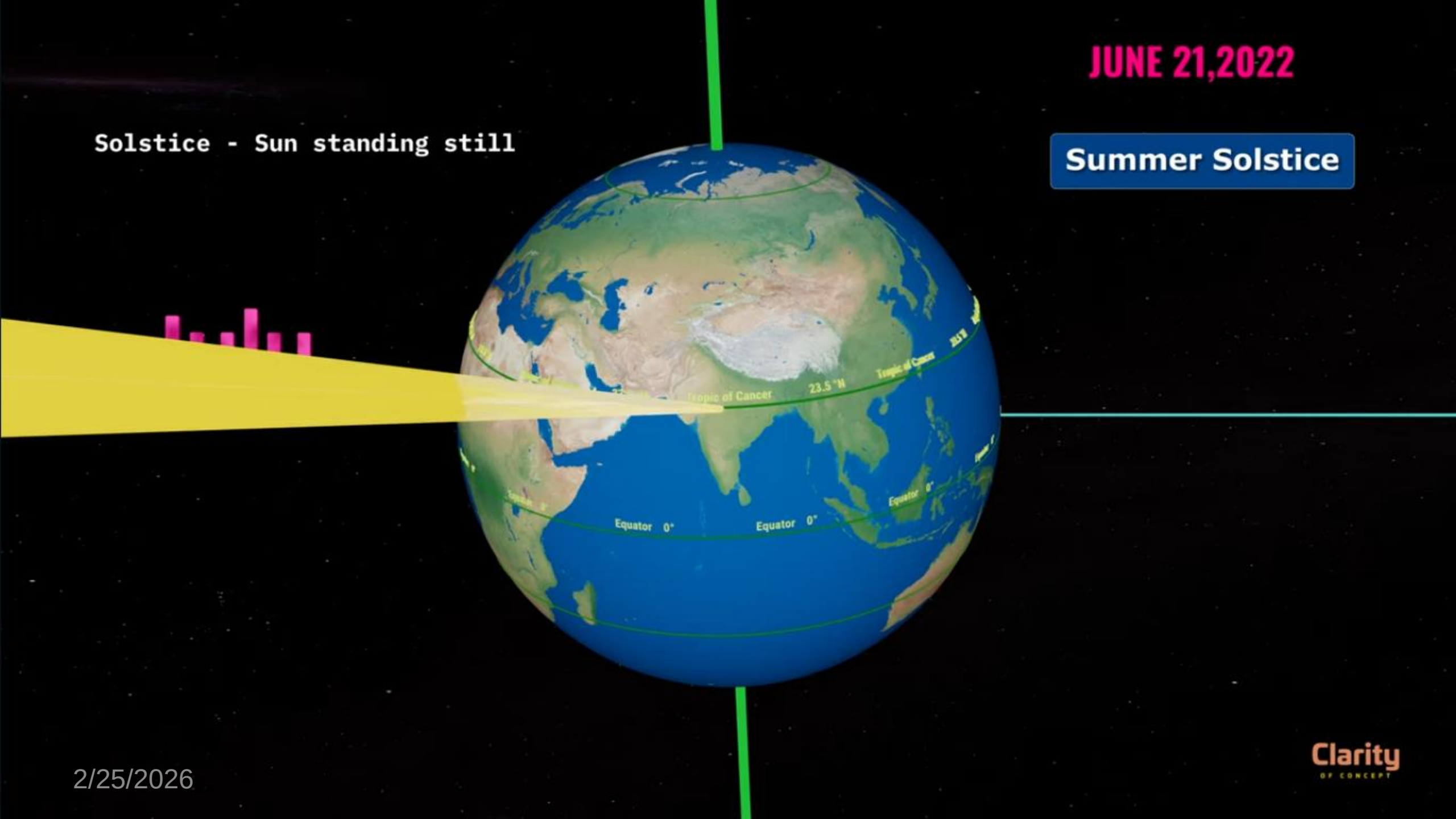
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Clarity
OF CONCEPT

JUNE 21, 2022

Solstice - Sun standing still

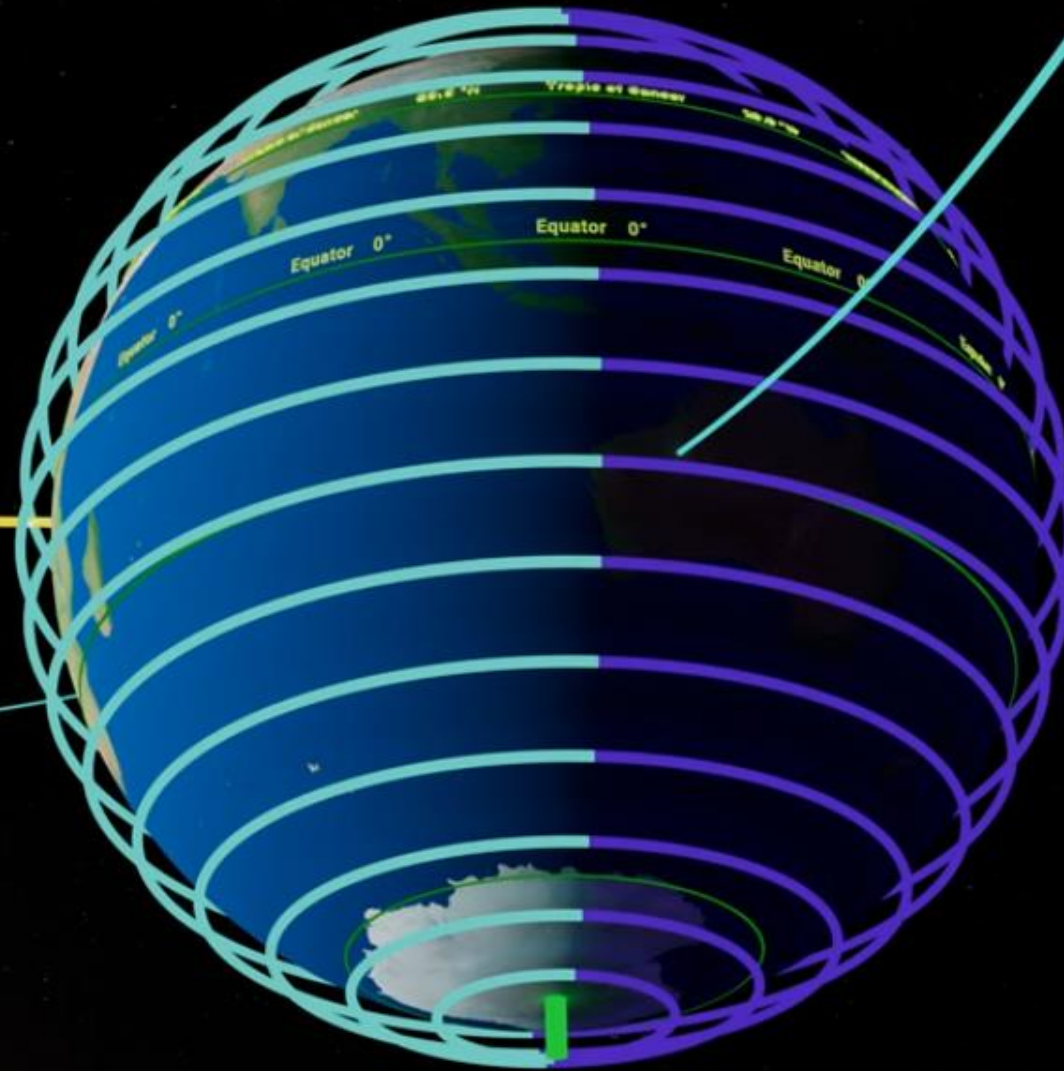
Summer Solstice



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Clarity
OF CONCEPT

SEPTEMBER 23, 2022



November

After Summer
and before Winter,
the season is
known as Autumn..

Equinox -
Equal night

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DECEMBER 22, 2022

Winter Solstice



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Clarity
OF CONCEPT

Dynamical Definition Of The Tropics

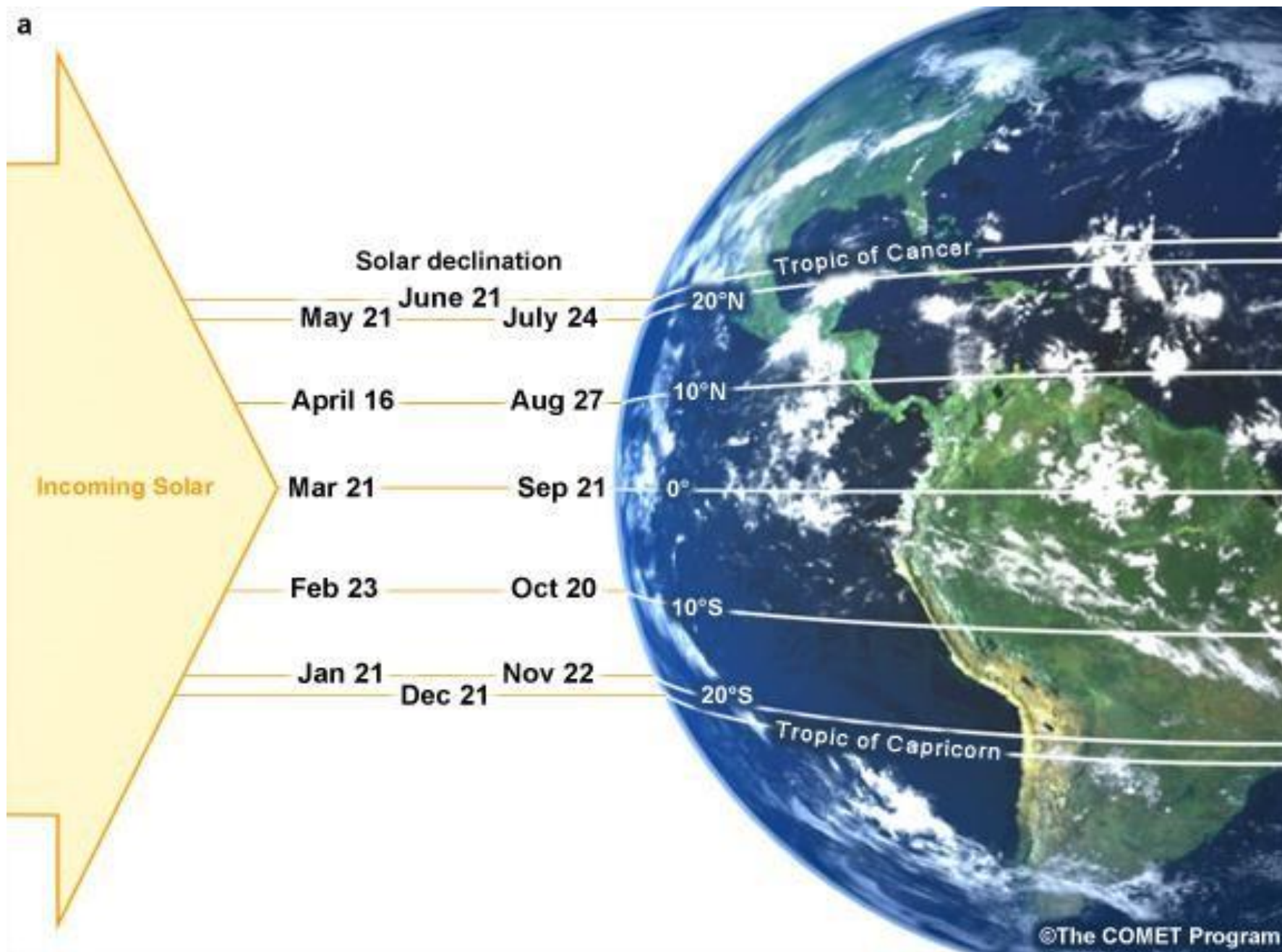
- Meteorologically, the tropics are often considered as
30°N to 30°S

Why wider?

- Because major tropical atmospheric circulations extend beyond 23.5°, including:
 - ✓ The Hadley circulation
 - ✓ Trade winds
 - ✓ Tropical cyclones
 - ✓ ITCZ migration

- This definition is more useful for atmospheric dynamics studies.

✱ Tropics can be defined as a region of surplus radiation with net upward motion, easterly boundary layer flow and lower surface pressures associated with a meridional overturning upper level circulation.



Climatic Boundaries Of The Tropics

Climatic Boundaries of the Tropics

- The climatic boundary is determined by:
 - Mean temperature patterns
 - Radiation surplus
 - General circulation structure
 - Rainfall distribution
 - Pressure belts

Climatic Boundaries Of The Tropics

- The climatic boundaries of the tropics includes the climate in Tropic of Cancer and the Tropic of Capricorn(30° N and 30° S).
- Tropics is a warm region; mean annual temperatures exceed 20°C almost throughout the Tropics, except at high elevations, and exceed 25°C in many parts of the tropical zone.
- The tropical zone encompasses some of the wettest locations on Earth, as well as some of the world's driest deserts.

Cont.....

- The Tropics include the driest parts of the world, the Atacama Desert on the west coast of South America where mean annual rainfall is less than 1 mm, as well as some of the wettest, locations in western Colombia and in Hawaii which receive more than 10,000 mm a year.
- High-altitude areas within the Tropics, mean temperatures typically decrease at a rate of about 6°C per 1000 meters.

Main Characteristics Of The Tropical Weather And Climate

Main Characteristics Of The Tropical Weather

The tropical atmosphere is primarily controlled by:

- High solar radiation
- Weak Coriolis force
- Abundant moisture
- Large-scale overturning circulation
(Hadley circulation)

Main Characteristics Of The Tropical Weather

- Tropical weather and climate are fundamentally different from those of mid- and high latitudes.
- While mid-latitudes are dominated by **strong** horizontal temperature gradients and frontal systems, the tropics are dominated by vertical motion, convection, and latent heat release.
- High-altitude areas within the Tropics, mean temperatures typically decrease at a rate of about 6°C per 1000 meters.

Main Characteristics Of The Tropical Weather And Climate

- ❖ Tropical regions are characterized by **high temperatures** throughout the year due to their location near the equator.
- ❖ These areas experience **abundant sunshine** and receive significant amounts of precipitation, often in the form of convectional rain.
- ❖ Tropical climates are generally divided into three main types: tropical rainforest, tropical monsoon, and tropical savanna.
- ❖ Weather patterns in these regions are influenced by phenomena like the Intertropical Convergence Zone (ITCZ), trade winds, and tropical cyclones.
- ❖ Humidity is typically high, contributing to the frequent occurrence of thunderstorms.

Main Characteristics Of The Tropical Weather And Climate

- The classical definition of a tropical climate is one in which the mean temperature of the coolest month exceeds 18°C .
- A small annual range of temperature is characteristic of the Tropics. Near the Equator, the difference in mean temperature between the warmest and coldest months is rarely more than 4°C , and at some locations such as Quito, Ecuador it is less than 1°C .
- Annual ranges of temperature become larger as one moves further from the Equator, but even at the limits of the tropical zone, few locations have an annual range of more than 15°C .

- One of the main features of tropical regions is the very high absorption of heat and its low emanation.
- abundant rainfall also A key characteristic of tropical weather.

Tropical Meteorology



Chapter 1: Introduction



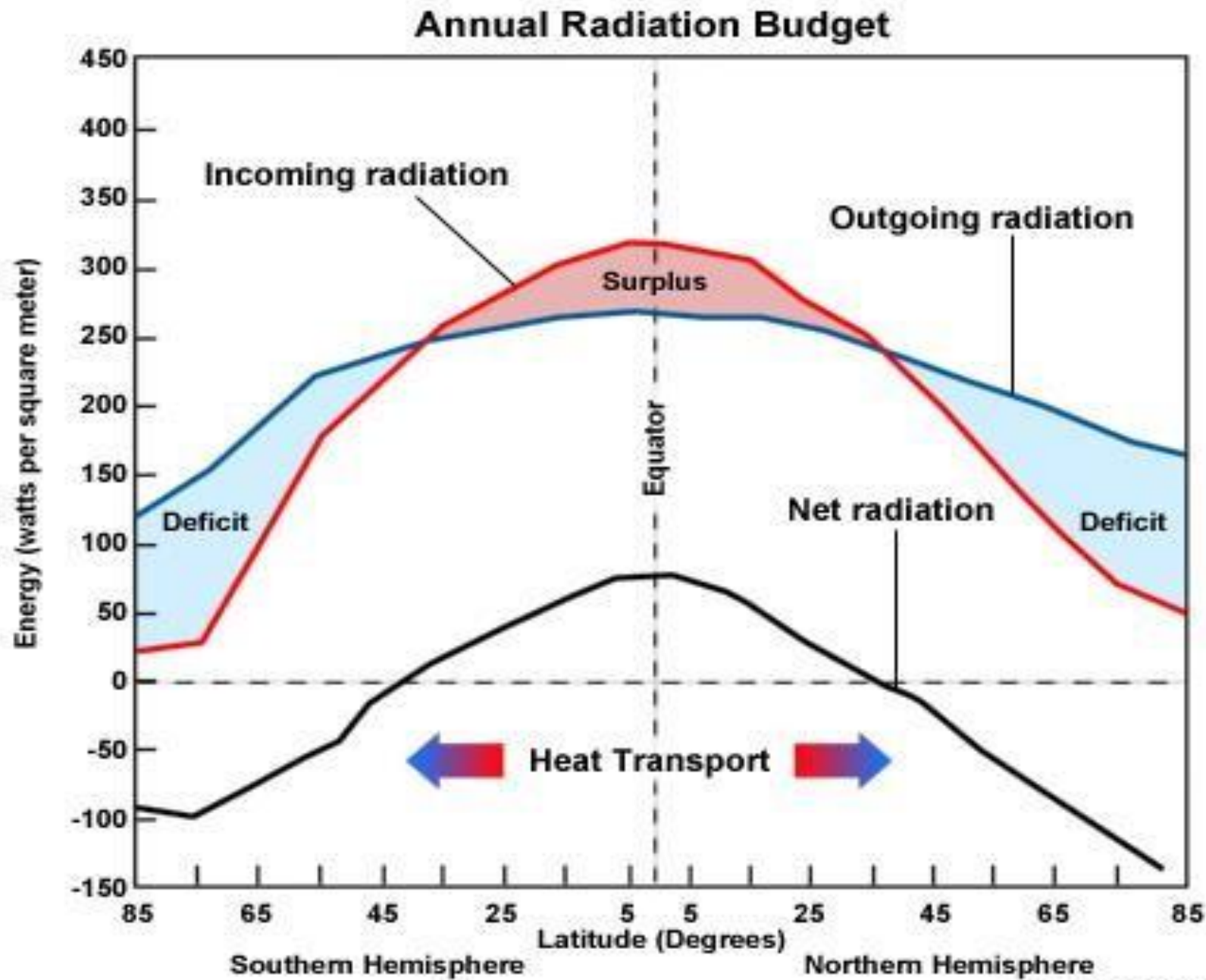
Objective

- *To Describe The Development Of Convection In The Tropics*
- *To Know Importance Of Tropics In Atmospheric General Circulation*
- *To Know The Significant Of Studying Tropical Meteorology*

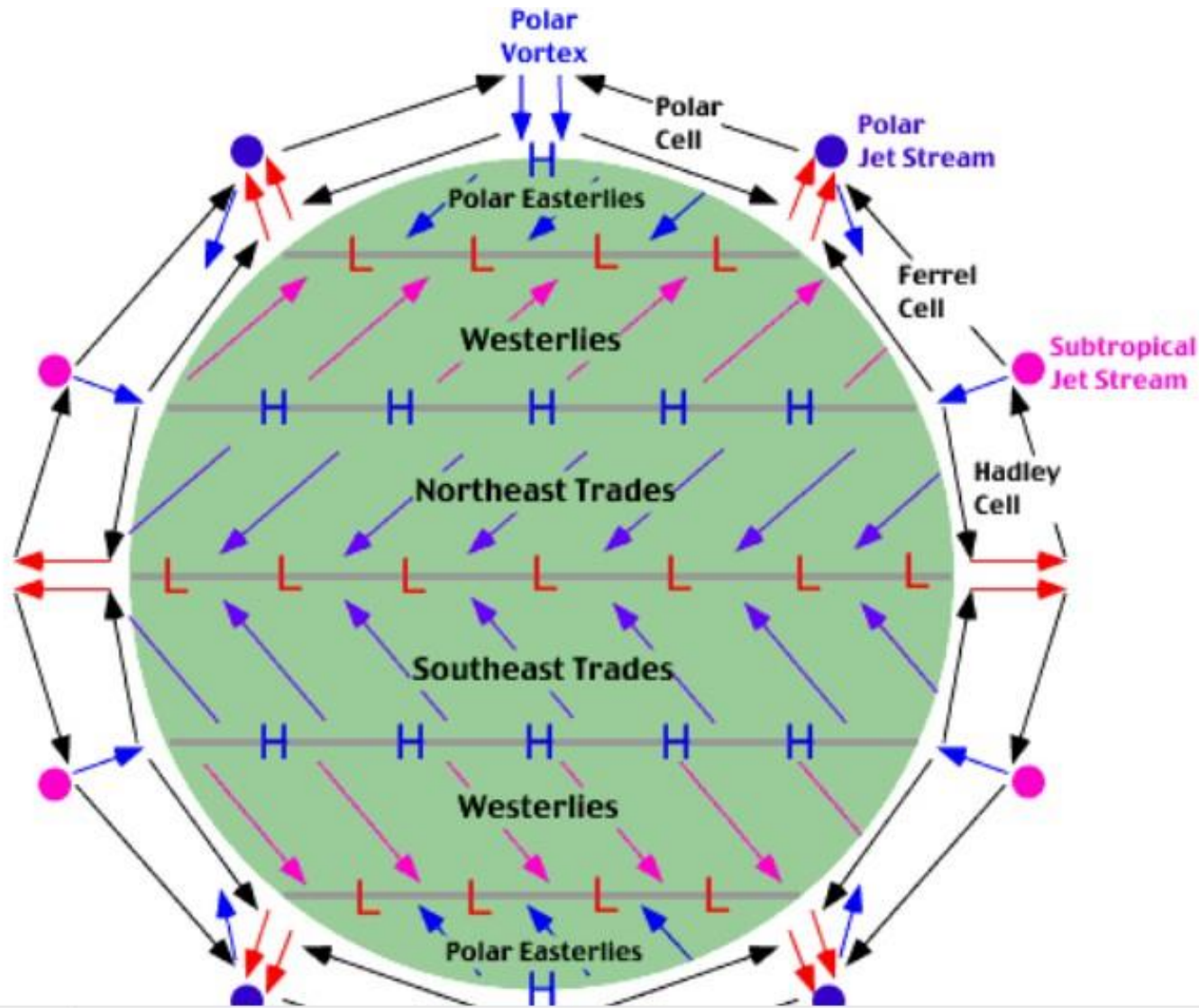
Importance Of Tropics In Atmospheric General Circulation

- The annually averaged incoming solar radiation per unit area of the earth's surface is much greater at the equator than at the poles
- A difference that is compounded by the fact that the polar regions are covered in ice and snow and therefore reflect much of the incoming radiation back to space.
- A little less obvious is the fact that the tropical regions actually receive more energy from the sun than they emit back to space, while the converse is true in high latitudes.
- *Since both regions are, on an annual average, in equilibrium, there must be a process acting to transport excess energy from the tropics to make up the deficit in high latitudes.*

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- The most fundamental feature of the tropical atmospheric circulation is a zone where warm (less dense) air rises from the surface to the upper atmosphere (a process known as **convection**), creating a zone of low pressure at the surface.
- Once in the upper atmosphere, this air moves towards the poles before descending in the subtropics, the descending air creating a zone of high surface pressure (the subtropical high or ridge).
- To complete the process, air then flows at surface level from the high-pressure area in the subtropics towards the low-pressure zone near the Equator. The whole structure is known as the **Hadley Cells**.



➤ The northern boundary of these cells is marked by strong westerly winds in the upper troposphere called the **tropospheric jets**.

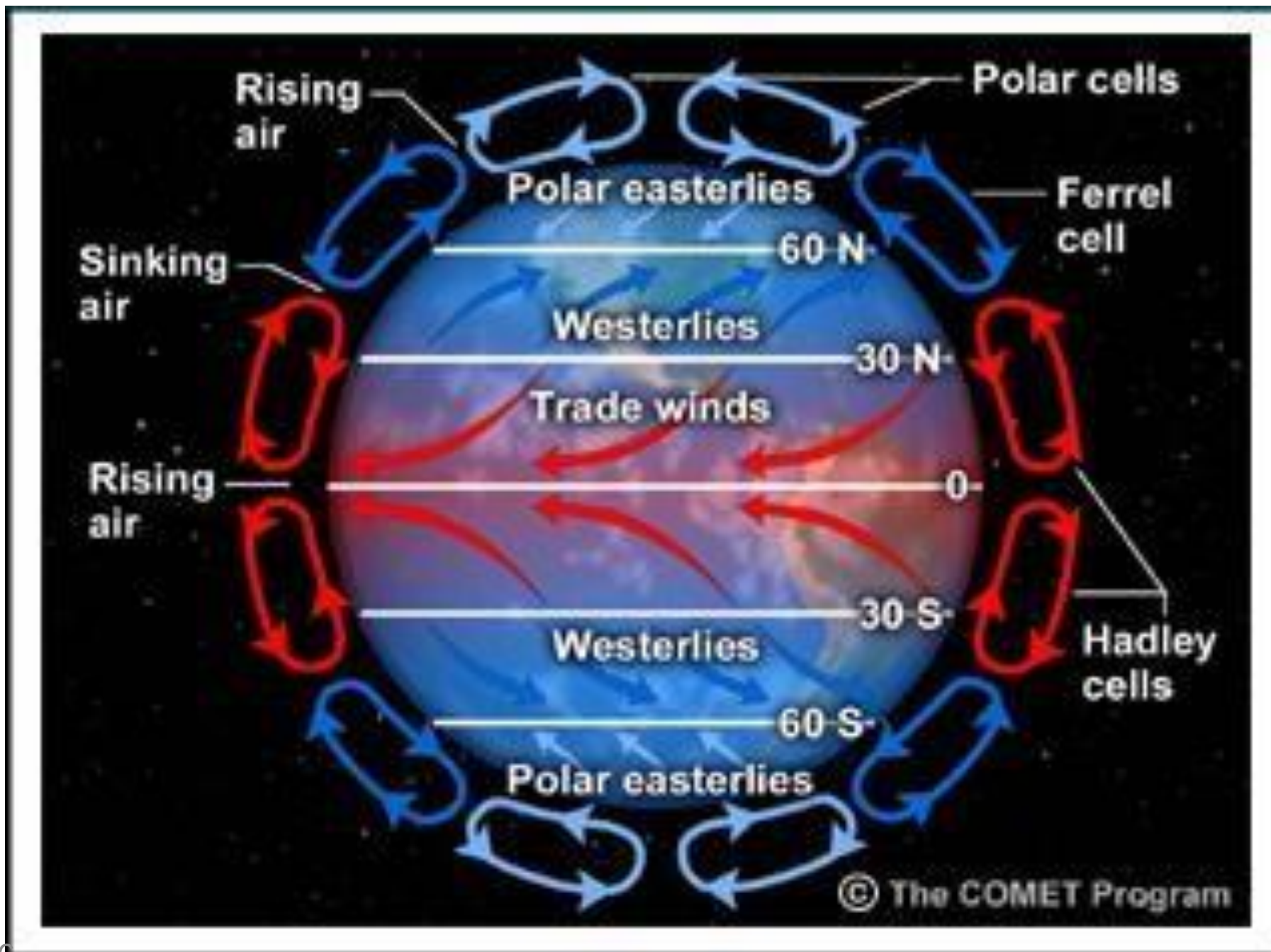
➤ At the surface, the Earth's rotation is responsible for a deflection toward the right in the northern hemisphere and toward the left in the southern hemisphere (due to the **Coriolis force**) of the flow coming from the mid-latitudes to the Equator. This gives rise to the easterly **trade winds** characteristics of the tropical regions.

- ❖ Vertical motion leads to adiabatic temperature change, thus the location of high and low pressures give us information about expected cloudiness and precipitation.
- ❖ Rising motion in low pressure leads to expansion, adiabatic cooling, and condensation while sinking in high pressures leads to adiabatic compression, warming and dry conditions.

General Circulation of the Atmosphere

- The major influences on the general circulation of the atmosphere are:
 - Differential heating
 - Rotation of the planet
 - Topography
 - Atmospheric and oceanic fluid dynamics
- the general circulation for an aqua planet is constituted with tropical easterly surface winds, midlatitude westerlies, and polar easterlies
- When the impact of continents is added, more complex surface patterns develop; Instead of bands of low and high pressure, areas of high and low pressure form in different regions, e.g., subtropical highs are centered over the ocean.

- 🌺 The tropics are marked by upward motion and the subtropics by sinking motion, forming the Hadley cell.
- 🌺 The maximum upward motion occurs during the summer over the more continental Northern Hemisphere.
- 🌺 The pattern of rising motion flanked by dual areas of sinking motion occurs only during the southern hemisphere summer.
- 🌺 In the north, the continental effects lead to strong upward motion in the lower troposphere during summer and the strong sinking motion during winter.



The Significant Of Studying Tropical Meteorology?

Why Tropics?

- Tropical climates are of especial geographical interest because 50 per cent of the surface of the globe lies between latitudes 30°N and 30°S, and over 75 per cent of the world's population inhabit climatically tropical lands.

Why Tropics?

- The ability of tropical societies to reduce their vulnerability to environmental disasters is tied to the quality and application of weather and climate information and prediction.
- For example, knowledge of the spatial and temporal variability of precipitation is needed to manage agriculture, water resources, public health, and renewable energy.
- Such information is gained from analysis of the observations and predictive models.

Why Tropics?

- Advances in tropical prediction are due to the combination of physical insight, skillful computer model representations of the physical laws governing the weather and climate, and advances in computing technology.
- Numerical (computer) models also facilitate examination of physical processes that influence weather and climate.
- Knowledge of the tropical atmosphere has increased significantly during recent decades, especially because of satellite observations, field programs, and modern numerical models.

Why Do Tropical Regions Have Both Deserts and Rainforests?

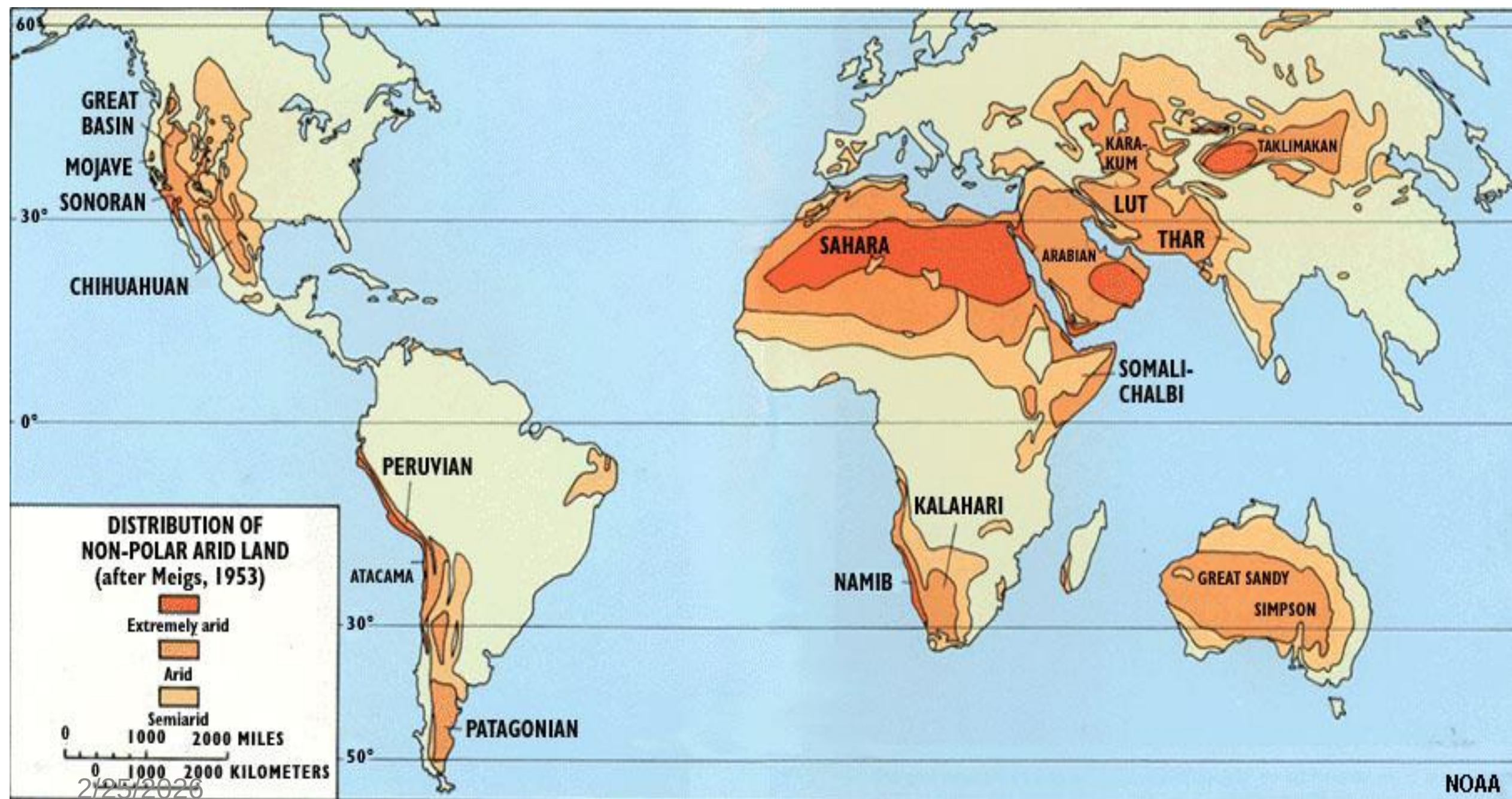
The Role of the Hadley Circulation

The Hadley circulation has two main vertical branches:

- Rising air near the Equator
- Sinking air near 25° – 35° latitude

These two motions create completely different climates.

- Tropical rainforests exist where air rises and produces heavy rainfall, while tropical deserts exist where air sinks and suppresses rainfall.
- The Hadley circulation creates both environments within the tropical and subtropical belt.

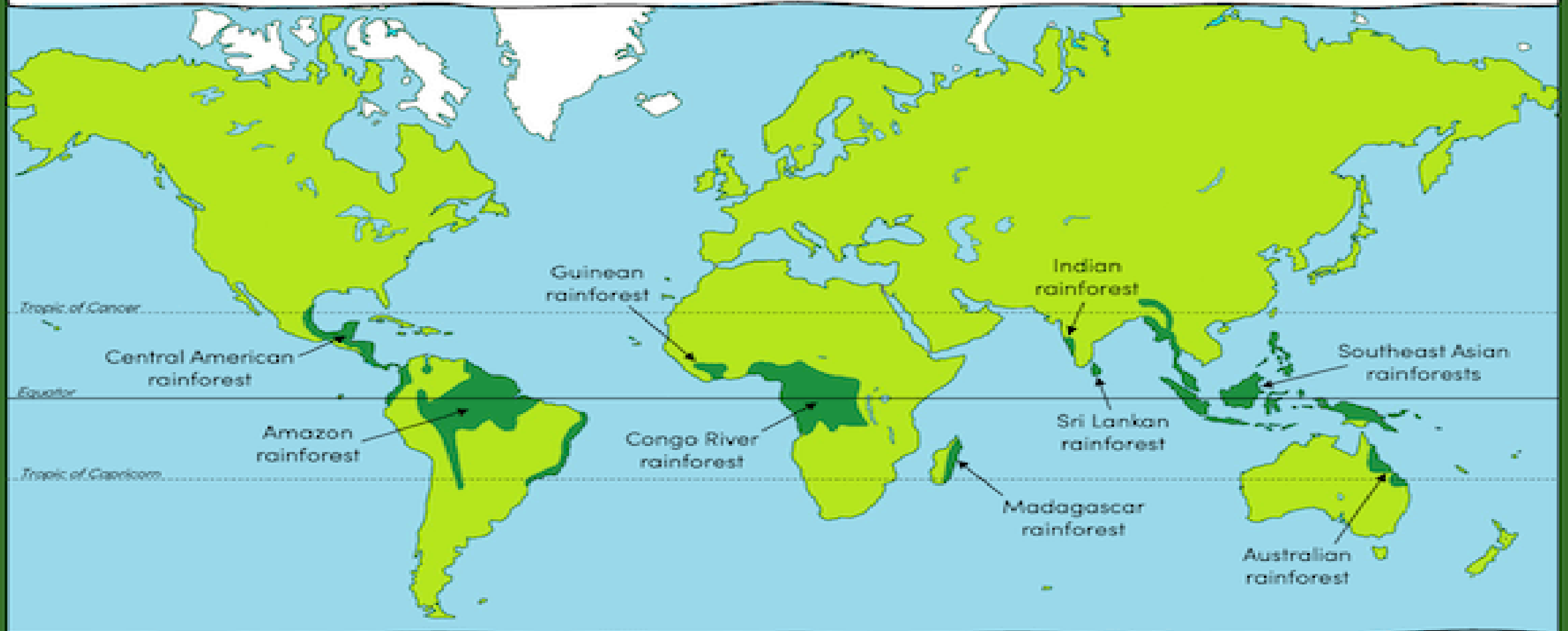


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Tropical rainforests are mainly found in the Earth's hot, sunny tropical zone.



High temperatures make the air very humid (warm and wet) in many parts of the tropics, so it rains - a lot!



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Thanks !